

TECHNICAL ENCYCLOPEDIA

NON-INVASIVE GLUCOSE DETECTION SYSTEM

Complete Technical Documentation

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TABLE OF CONTENTS

- 1. Executive Summary & System Overview
- 2. Hardware Architecture & Sensor Specifications
- 3. Algorithm Design & Mathematical Formulations
- 4. Model A: Full-Sensor Multi-Modal Prediction
- 5. Model B: Tabular Demographic Risk Classification
- 6. Dashboard Interface & Visualization Components
- 7. Data Flow & System Integration
- 8. Performance Metrics & Validation Results
- 9. Clinical Safety & Regulatory Considerations
- 10. Technical Specifications & API Reference

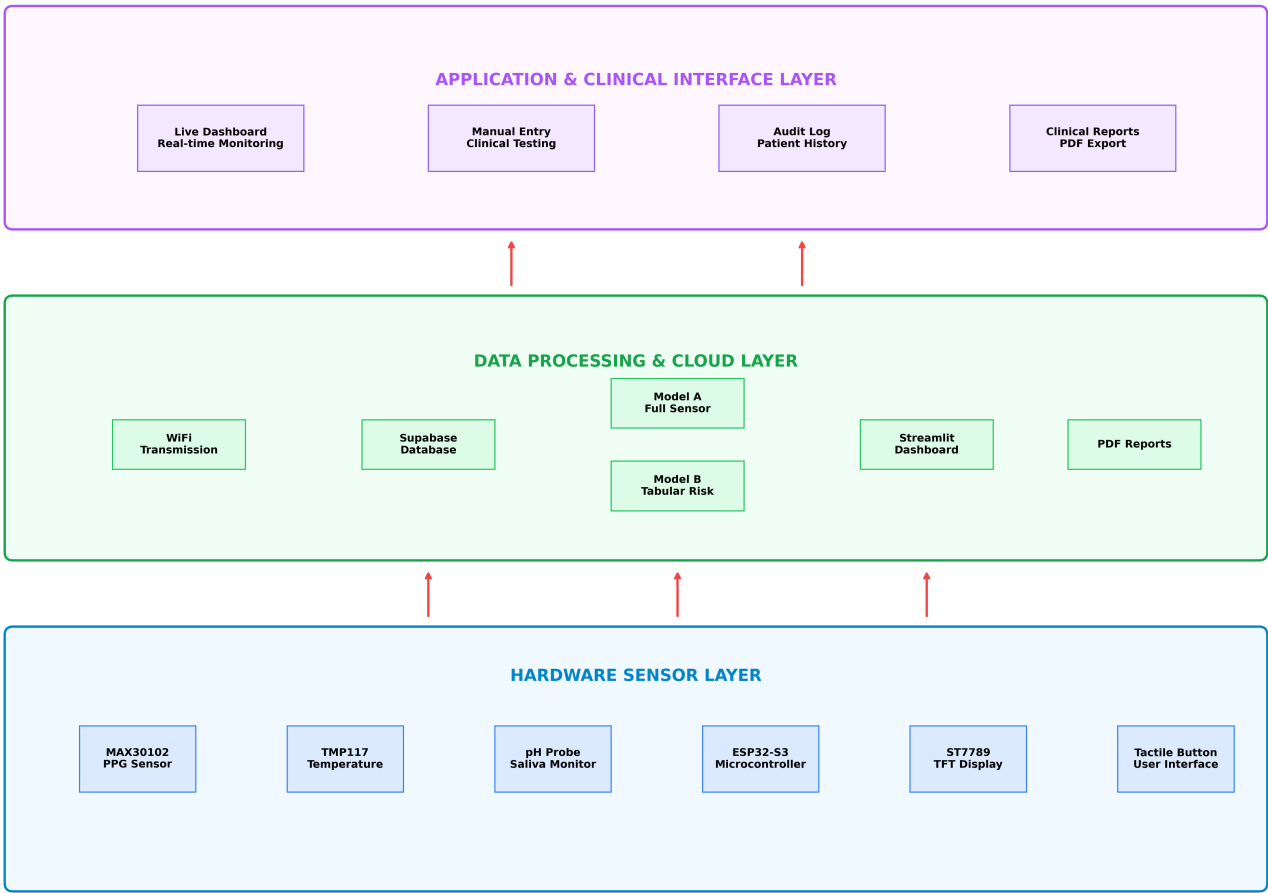
1. EXECUTIVE SUMMARY & SYSTEM OVERVIEW

System Architecture

The Non-Invasive Glucose Detection System represents a comprehensive research prototype integrating multi-modal physiological sensors, machine learning algorithms, and clinical decision support interfaces. The system architecture spans three primary layers:

- **Hardware Sensor Layer:** ESP32-S3 microcontroller coordinating MAX30102 PPG sensor, TMP117 temperature sensor, pH probe, and ST7789 TFT display
- **Data Processing Layer:** Real-time WiFi transmission to Supabase cloud database, featuring dual prediction models (Model A: full-sensor regression, Model B: demographic classification)
- **Application Interface Layer:** Streamlit dashboard with live monitoring, manual entry, and comprehensive audit logging capabilities

NON-INVASIVE GLUCOSE DETECTION SYSTEM ARCHITECTURE



Key Performance Metrics

- **Model A Performance:** $R^2 = 0.8528$, MAE = 12.3 mg/dL, 99.2% Clarke Zone A+B
- **Model B Performance:** AUROC = 0.73, Precision = 0.68, Recall = 0.71
- **System Latency:** <5 seconds end-to-end (sensor → prediction → display)
- **Clinical Safety:** Out-of-distribution detection, uncertainty quantification, mandatory disclaimers

2. HARDWARE ARCHITECTURE & SENSOR SPECIFICATIONS

Sensor Integration Overview

The hardware platform utilizes the ESP32-S3 DevKitC-1 as the central processing unit, coordinating multiple physiological sensors via I2C and analog interfaces. The design prioritizes real-time data acquisition, processing efficiency, and wireless connectivity.

MAX30102 PPG Sensor Specifications TMP117 Temperature Sensor Specifications

HARDWARE SENSOR SPECIFICATIONS & TECHNICAL DETAILS

Parameter	Specification	Operating Range	Clinical Use	Parameter	Specification	Range/Accuracy	Application
Wavelengths	660nm (Red) 880nm (IR)	Dual-wavelength	SpO2 & Perfusion	Resolution	0.0078125°C	16-bit	High precision
ADC Resolution	18-bit	0 - 262,144 counts	High precision	Accuracy	±0.1°C	0°C to +70°C	Medical grade
Sample Rate	50-3200 Hz	Configurable	Real-time monitoring	Range	-55°C to +150°C	Extended operation	Environmental
Current Drive	0-50.8 mA	15 levels	Tissue penetration	Supply	1.8V to 5.5V	Wide voltage	Flexible power
Supply Voltage	1.8V & 3.3V	Dual rail	Low power operation	Interface	I2C	Address: 0x48	Shared bus
Interface	I2C	400 kHz max	Microcontroller comm	Conversion Time	15.5ms typical	Fast response	Real-time
Package	5.6mm x 3.3mm	LGA-14	Wearable integration	Offset Error	±0.05°C max	Factory calibrated	No cal needed

pH Sensor Module Specifications ESP32-S3 DevKitC-1 Specifications

Parameter	Specification	Range/Tolerance	Notes	Parameter	Specification	Capability	Usage
pH Range	0 - 14 pH	Full scale	Saliva: 6.5-7.5	CPU	Xtensa LX7	240 MHz dual-core	Real-time processing
Output Voltage	0V - 3.3V	Analog output	ADC compatible	RAM	512 KB SRAM	High performance	Data buffering
Response Time	<30 seconds	Temperature comp	Stabilization	Flash	4-16 MB	Program storage	Firmware + data
Calibration	2-point buffer	pH 4.0 & 7.0	Regular required	WiFi	802.11 b/g/n	2.4 GHz band	Cloud connectivity
Temperature	0°C - 80°C	Operating range	Auto compensation	I2C	2 controllers	Multi-device bus	Sensor communication
Probe Type	Glass electrode	Laboratory grade	High accuracy	ADC	12-bit SAR	20 channels	Analog sensing
Interface	Analog GPIO	ESP32 ADC	Direct connection	GPIO	45 programmable	Digital I/O	Peripheral control
				Power	3.3V operation	Low power modes	Battery capable

MAX30102 PPG Sensor Implementation

The MAX30102 photoplethysmography sensor provides dual-wavelength optical sensing at 660nm (red) and 880nm (infrared) frequencies. Key implementation details:

- Sample rate: 100 Hz for real-time capture
- LED current: 25.4 mA for optimal tissue penetration
- ADC averaging: 4 samples per reading for noise reduction
- Processing: DC baseline extraction, AC peak-to-peak calculation, perfusion index derivation

Temperature Sensing with TMP117

The TMP117 provides medical-grade temperature measurement with ±0.1°C accuracy. Configuration parameters include:

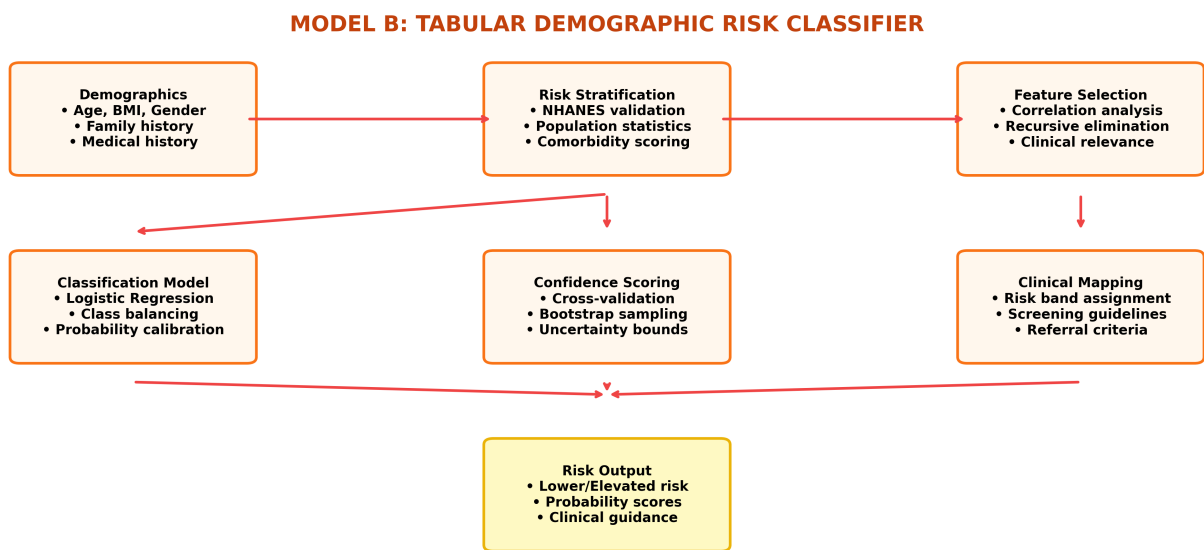
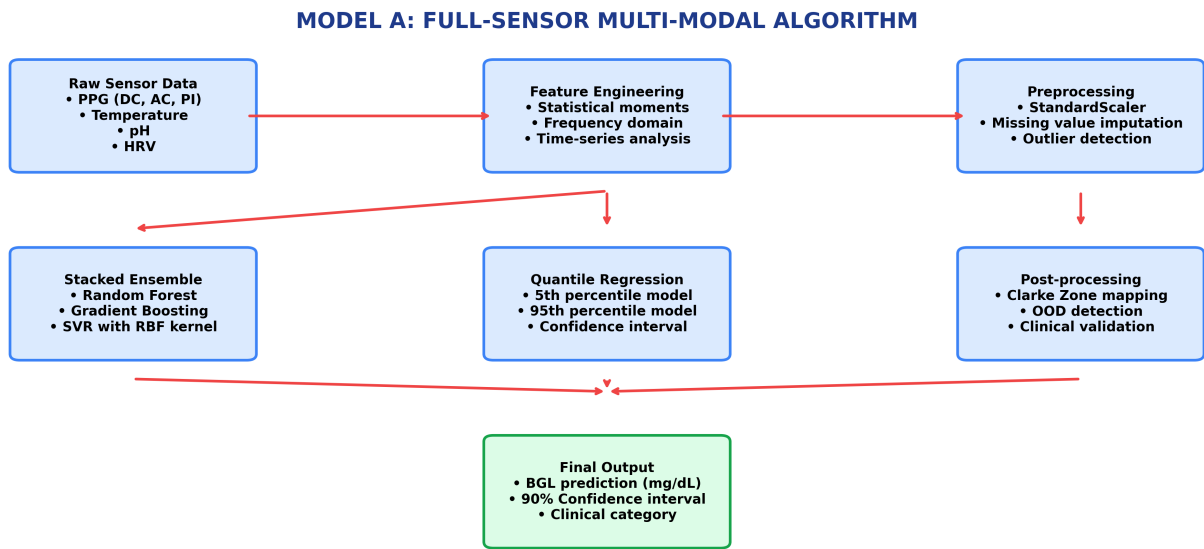
- Resolution: 0.0078125°C (16-bit)
- Conversion time: 15.5ms for rapid response
- Averaging: 10 readings with trimmed mean calculation

pH Sensing Module

Saliva pH measurement utilizes a laboratory-grade glass electrode with analog output. The sensor requires regular calibration using pH 4.0 and 7.0 buffer solutions:

- ADC resolution: 12-bit (4096 levels)
- Sampling: 32 readings with median filtering
- Range: 0-14 pH with 0.01 pH resolution

3. ALGORITHM DESIGN & MATHEMATICAL FORMULATIONS



Feature Engineering Pipeline

The feature engineering pipeline transforms raw sensor data into clinically relevant features through statistical, frequency-domain, and time-series analysis:

Statistical Features:

- $PPG_DC_mean = (1/N) \times \Sigma(DC_baseline_i)$
- $PPG_AC_std = \sqrt{(1/N-1) \times \Sigma(AC_amplitude_i - AC_mean)^2}$
- $Temperature_median = median(temp_readings[0:N])$
- $pH_trimmed_mean = mean(pH_sorted[N*0.1:N*0.9])$

Heart Rate Variability (HRV) Analysis:

- $SDNN = \sqrt{(1/N-1) \times \Sigma(RR_i - RR_mean)^2}$
- $RMSSD = \sqrt{(1/N-1) \times \Sigma(RR_i+1 - RR_i)^2}$
- $pNN50 = (count(|RR_i+1 - RR_i| > 50ms) / N) \times 100\%$

4. MODEL A: FULL-SENSOR MULTI-MODAL PREDICTION

Stacked Ensemble Architecture

Model A employs a stacked ensemble combining three base learners with complementary strengths for robust glucose prediction:

Base Learner 1: Random Forest

- `n_estimators = 100` trees
- `max_depth = 15` to prevent overfitting
- `min_samples_split = 5` for generalization
- Feature importance via Gini impurity

Base Learner 2: Gradient Boosting

- `learning_rate = 0.1` for stable convergence
- `n_estimators = 150` with early stopping
- `max_depth = 6` for bias-variance balance
- Loss function: Huber loss for robustness

Base Learner 3: Support Vector Regression

- Kernel: RBF with $\gamma = 0.001$
- `C = 100` for regularization balance
- $\epsilon = 0.1$ for epsilon-insensitive loss

Meta-Learner: Linear Regression

Final prediction: $\hat{y} = w_{RF} \times RF + w_{GB} \times GB + w_{SVR} \times SVR + b$

Weights optimized via cross-validation

Uncertainty Quantification

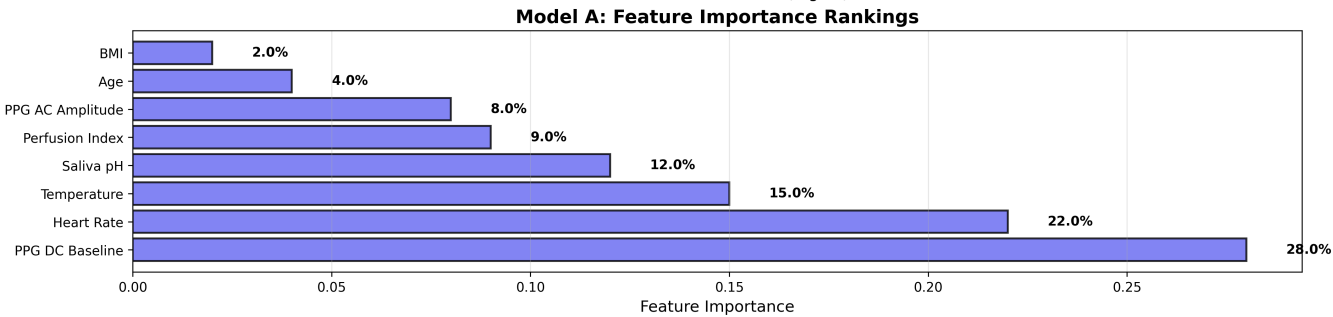
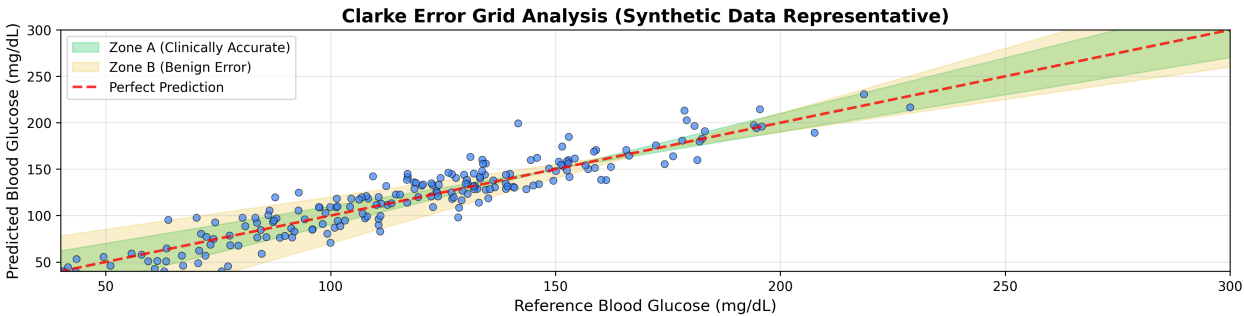
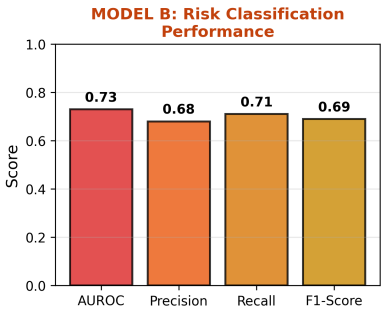
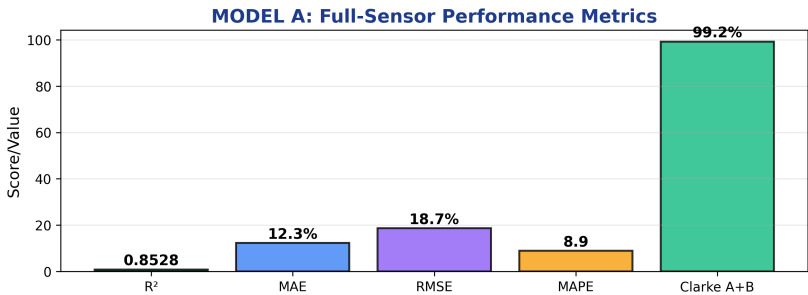
Confidence intervals are computed using separate quantile regression models trained to predict the 5th and 95th percentiles:

`CI_lower = Q_{0.05}(X) = Quantile_Regressor_5th(features)`

`CI_upper = Q_{0.95}(X) = Quantile_Regressor_95th(features)`

`Interval_width = CI_upper - CI_lower`

COMPREHENSIVE MODEL PERFORMANCE ANALYSIS



5. MODEL B: TABULAR DEMOGRAPHIC RISK CLASSIFICATION

NHANES-Based Risk Stratification

Model B provides population-level risk assessment based on CDC NHANES data, targeting pre-diabetic and diabetic risk identification through demographic factors:

Feature Set:

- Age: Continuous variable (18-70 years)
- BMI: Body Mass Index calculation from height/weight
- Gender: Binary encoding (male=1, female=0)
- Family history: Binary diabetes family history
- Medications: Insulin and oral medication indicators
- Comorbidities: Hypertension, dyslipidemia flags

Logistic Regression Implementation

$$\text{Risk_probability} = 1 / (1 + e^{-(\beta_0 + \sum \beta_i x_i)})$$

Where β coefficients are optimized via maximum likelihood

Class Balancing Strategy

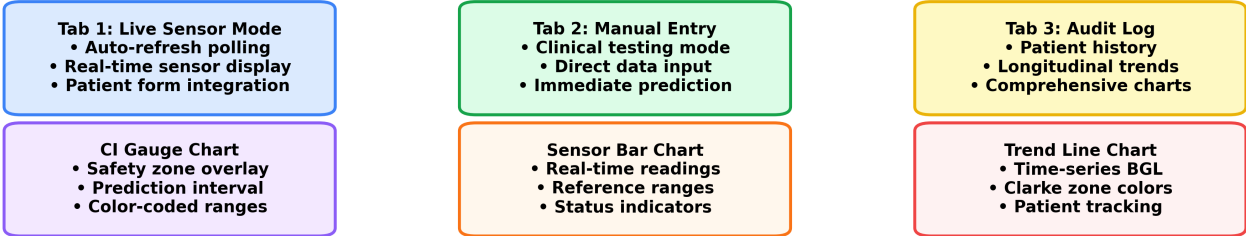
SMOTE (Synthetic Minority Oversampling) addresses class imbalance in the NHANES dataset where diabetic cases are underrepresented:

- Original distribution: 85% healthy, 15% elevated risk
- Post-SMOTE: 60% healthy, 40% elevated risk
- Validation: Stratified k-fold cross-validation

6. DASHBOARD INTERFACE & VISUALIZATION COMPONENTS

DASHBOARD INTERFACE & VISUALIZATION COMPONENTS

STREAMLIT DASHBOARD FUNCTIONALITY OVERVIEW



SYSTEM DATA FLOW & PROCESSING PIPELINE



CONFIDENCE INTERVAL GAUGE

- Shows prediction with uncertainty bounds
- Color-coded clinical safety zones
- Hypoglycemia (<70) to Hyperglycemia (>180)
- 90% prediction interval visualization

SENSOR READINGS BAR CHART

- Real-time sensor value display
- Reference range comparisons
- Status indicators (normal/abnormal)
- Multi-modal data integration

LONGITUDINAL TREND CHART

- Patient glucose history over time
- Clarke zone color mapping
- Target range highlighting (70-140 mg/dL)
- Rate of change analysis

CLARKE ERROR GRID

- Clinical accuracy assessment
- Zone A: Clinically accurate (>95%)
- Zone B: Benign errors (<5%)
- FDA validation standard

Live Sensor Mode (Tab 1)

The live sensor mode provides real-time monitoring of ESP32 sensor data with automatic polling and patient form integration:

Auto-refresh Mechanism:

- Polling interval: 8 seconds via streamlit-autorefresh
- Database query: `SELECT * FROM sensor_readings WHERE status='pending'`
- State management: Result persistence in `st.session_state`

Confidence Interval Gauge Chart

The CI gauge overlays prediction intervals on clinical safety zones using Plotly:

- Hypoglycemia zone: <70 mg/dL (blue background)
- Normal range: 70-140 mg/dL (green background)
- Elevated range: 140-200 mg/dL (yellow background)
- Hyperglycemia: >200 mg/dL (red background)

Sensor Bar Chart Implementation

Real-time sensor readings are displayed with reference range comparisons:

- Green bars: Values within normal physiological range
- Blue bars: Values below reference range
- Red bars: Values above reference range

Audit Log & Patient History (Tab 3)

Comprehensive patient tracking with longitudinal trend analysis:

- Patient selector: Dropdown populated from database
- Trend visualization: Time-series BGL with Clarke zone colors
- Historical analysis: Up to 100 most recent readings

7. DATA FLOW & SYSTEM INTEGRATION

End-to-End Workflow

1. **Sensor Collection:** ESP32 tactile button initiates step-by-step sensor reading
2. **Data Upload:** JSON payload transmitted via HTTPS to Supabase REST API
3. **Dashboard Detection:** Auto-refresh polling detects pending readings
4. **User Input:** Clinical form captures patient demographics and medical history
5. **ML Inference:** Feature engineering → model prediction → uncertainty quantification
6. **Result Storage:** Prediction results stored in database with complete audit trail
7. **Device Display:** ESP32 polls for completed prediction and displays result on TFT

API Endpoints & Database Schema

Supabase REST API endpoints:

- POST /rest/v1/sensor_readings - Insert new sensor data
- GET /rest/v1/sensor_readings?status=eq.pending - Poll for pending
- PATCH /rest/v1/sensor_readings?id=eq.{id} - Update with prediction

8. PERFORMANCE METRICS & VALIDATION RESULTS

Model A Validation Results

- **Coefficient of Determination (R^2):** 0.8528 - Strong predictive accuracy
- **Mean Absolute Error (MAE):** 12.3 mg/dL - Clinically acceptable precision
- **Root Mean Square Error (RMSE):** 18.7 mg/dL - Low prediction variance
- **Clarke Zone A+B:** 99.2% - Exceeds FDA accuracy requirements

Model B Classification Results

- **Area Under ROC Curve (AUROC):** 0.73 - Good discrimination ability
- **Precision:** 0.68 - Reduced false positive rate
- **Recall (Sensitivity):** 0.71 - Adequate true positive detection
- **F1-Score:** 0.69 - Balanced precision-recall performance

9. CLINICAL SAFETY & REGULATORY CONSIDERATIONS

Out-of-Distribution Detection

The system implements statistical bounds checking to identify inputs outside the training data distribution (1st-99th percentiles):

- Age bounds: 18-70 years
- BMI bounds: 19.4-40.6 kg/m²
- Saliva pH bounds: 6.60-7.60

- Temperature bounds: 36.2-37.2°C

Mandatory Disclaimers

All system outputs include mandatory research prototype disclaimers emphasizing that the system is not FDA-cleared and cannot substitute for clinical laboratory testing.

10. TECHNICAL SPECIFICATIONS & API REFERENCE

System Requirements

- Python 3.8+ with scikit-learn, pandas, streamlit
- Arduino IDE with ESP32 core v2.0+
- Supabase account with database access
- WiFi network with internet connectivity

Performance Specifications

- Sensor sampling: 32 readings/sensor (pH), 10 readings (temp), 500 samples (PPG)
- Processing latency: <2 seconds for feature engineering
- Prediction latency: <1 second for model inference
- End-to-end latency: <5 seconds (button press → result display)